Financial Ambassador to America and Europe in which capacity, he conducted the preliminary negotiations for a wheat loan from the US. Pt Nehru had also asked him to work on the organization of the Planning Commission and appointed him as a member when it was set up in April 1950. Thereafter he joined the Union Cabinet as Finance Minister till he resigned in July 1956 when Pt Nehru announced in a public meeting in Bombay that the city of Bombay would be centrally administered. The Government of India had reorganized the states on a linguistic basis, and instead of including the city of Bombay in the new state of Maharashtra there was a plan at that time to have a separate state of Bombay City. He made significant contributions to the formulation and implementation of the First and Second Five Year Plans. He was also primarily responsible for important landmarks such as the nationalization of the Imperial Bank of India, Life Insurance Companies, and the enactment of a new Companies Act.

After his resignation from the Union Cabinet, Deshmukh was active in the fields of education and social service. He was the Chairman of the University Grants Commission (1956–60), Vice-Chancellor of Delhi University (1962–67), President of the Indian Statistical Institute (1945–64), President of the Institute of Economic Growth (1965–74), and Chairman of the National Book Trust (1957–60). He was also founder of the India International Centre in 1959, Chairman of the Indian Institute of Public Administration (1963–64), and was the head of the Court of Governors of the Administrative Staff College of India (1953–73).

Deshmukh's first wife, Rosina, passed away in 1949 and in 1953 he married Durgabai, equally eminent in her own right. His contribution along with his wife in the field of functional literacy and family planning work undertaken by the Andhra Mahila Sabha, of which Durgabai was the founder President, was outstanding. Deshmukh and his wife were both awarded the Padma Vibhushan in 1975, the only instance of a husband-wife team being the recipient of India's second highest civilian honour. When Sir John Matthai came to know of the engagement, he told Deshmukh with a twinkle in his eye, 'that is a very powerful alliance.'[10] They had a pact: whenever one of them held a government post, the other would accept only ₹1 as salary. Unsurprisingly, Deshmukh was also the recipient of many other honours. He was a recipient of the Ramon Magsaysay Award for